

Mini-Project #1: Propensity Score Matching

Stat 437 — Right Heart Catheterization (RHC)

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1. Introduction

Welcome to a major challenge in matching. You are analyzing data from a study on the **Right Heart Catheterization (RHC)**, a diagnostic procedure used for critically ill patients. This is a legitimately challenging dataset to work with. Look [here](#) for more information about the dataset.

The Causal Problem: In the raw data, patients who receive RHC die at higher rates than those who don't. Does the procedure kill people? Or are doctors only performing the procedure on the patients who are already most likely to die?

Your goal in this analysis is to estimate the ATT of RHC on `survival`. Only include variables that are measured prior to RHC decision.

The Data

- **Treatment (RHC):** 1 (Treated) or 0 (Control).
- **Outcome (survival):** Whether the patient survived (1 = yes/0 = no).
- **Confounders:** This dataset is “messy.” It includes age, income, insurance type, and various physiological markers (mean blood pressure, heart rate, etc.).

2. Preliminary Data Inspection

We will load the data directly from `ATbounds` package.

```
# Load the dataset
data(RHC)

# Restrict to a subset of variables + Treatment and Outcome
target_vars <- c("survival", "RHC", "age", "edu", "income1", "income2",
  "ninsclas_No_insurance", "aps1", "surv2md1", "meanbp1",
  "pafi1", "crea1", "scoma1", "dnr1_Yes", "cardiohx", "das2d3pc",
  "chfhx", "seps_Yes", "cat1_MOSF_Sepsis", "cat1_CHF",
  "ca_Yes", "sex_Female", "race_black", "wtkilo1", "temp1")

RHC_use <- RHC[, target_vars]

# View variables
# View(RHC_use)
```

3. Calculate the naive comparison of means between survival and RHC.

Answer Here

```
RHC_use %>%
  group_by(RHC) %>%
  summarize(
    survival_rate = mean(survival)
  )
```

```
# A tibble: 2 x 2
  RHC survival_rate
<dbl>         <dbl>
1     0         0.370
2     1         0.320
```

```
with(RHC_use,
  mean(survival[RHC == 1]) - mean(survival[RHC == 0]))
```

```
[1] -0.05072115
```

The naive comparison shows that patients who receive RHC have a lower survival rate than those who did not. This difference is quite small, and the outcome could be potentially confounded by the statement that those who are more likely to receive the RHC treatment would most likely be sicker.

4. Two plots showing the selection bias

Here you need to include two plots that show that the probability of RHC depends on some baseline variables. Show how the probability of RHC varies as a function of at least two baseline severity markers.

15 Recommended Variables: These represent a mix of baseline health and acute crisis markers. Exploring these will help you realize that the RHC group is significantly sicker than the control group. Compare the rates of RHC and **survival** against these variables.

1. **aps1** (APACHE Score): A general measure of ICU severity. Higher scores mean the patient is closer to death.
2. **surv2mdl1**: The physician's estimated 2-month survival probability at baseline. This is a direct measure of how "doomed" the doctor thought the patient was.
3. **age**: Older patients might be treated differently than younger ones.
4. **meanbp1**: Mean blood pressure. Low blood pressure often triggers aggressive intervention.
5. **paf1**: Oxygenation index (PaO2/FiO2 ratio). A key marker for respiratory failure.
6. **das2d3pc**: Duke Activity Status Index. Measures how functional the patient was before getting sick.
7. **scoma1**: Glasgow Coma Score component. Measures neurological status.
8. **creal**: Creatinine levels. High levels indicate kidney failure.
9. **dnr1_Yes**: Does the patient have a "Do Not Resuscitate" order on day 1? This heavily influences treatment intensity.
10. **cat1_MOSF_Sepsis**: Multi-organ system failure with sepsis. These are extremely high-risk patients.
11. **cardiohx**: Presence of a prior cardiac history.
12. **chfhx**: History of Congestive Heart Failure.
13. **ninsclas_No_insurance**: Does insurance status impact the choice to perform an expensive procedure?
14. **wtkilo1**: Weight in kilograms.
15. **temp1**: Body temperature.

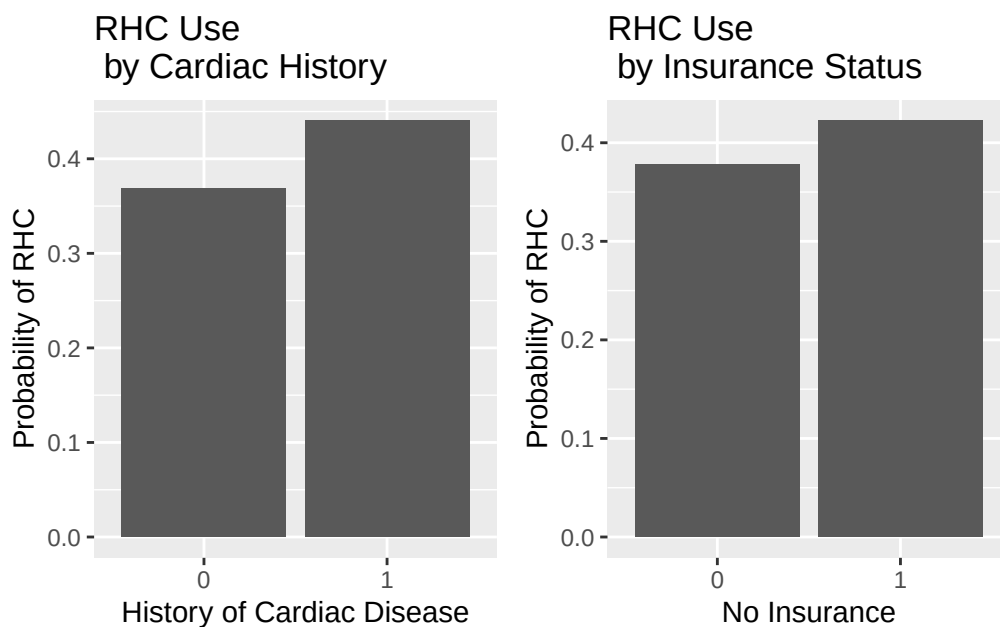
```
# put plots in here.  
library(patchwork)  
p1 <-ggplot(RHC_use %>%
```

```

    group_by(cardiohx) %>%
    summarize(prob_RHC = mean(RHC, na.rm = TRUE)),
    aes(x = factor(cardiohx), y = prob_RHC)) +
geom_col() +
labs(
  x = "History of Cardiac Disease",
  y = "Probability of RHC",
  title = "RHC Use \n by Cardiac History"
)

p2 <- ggplot(RHC_use %>%
  group_by(ninsclas_No_insurance) %>%
  summarize(prob_RHC = mean(RHC, na.rm = TRUE)),
  aes(x = factor(ninsclas_No_insurance), y = prob_RHC)) +
geom_col() +
labs(
  x = "No Insurance",
  y = "Probability of RHC",
  title = "RHC Use \n by Insurance Status"
)
p1+p2

```



RHC increases in both cases that there is a previous history of cardiac history, and when the

patient when insurance status changes. Both show in part that RHC is not randomly assigned, but rather is influenced by non-clinical characteristics. Both of the covariates are binary, so the comparison is between the two groups.

5. The Propensity Score Model

Task: Define your `ps_formula`. You have dozens of variables (e.g., `age`, `edu`, `cat1`, `das2d3pc`, `meanbp1`).

Justification: Given how “sick” the treated group is, which physiological markers are essential to include? Did you use any squared terms or interactions to improve the prediction of who gets an RHC?

Add Justification Here Core severity markers such as `aps1`, `surv2md1`, `meanbp1`, `pafi1`, `scoma1`, `crea1`, `seps_Yes`, `cat1_MOSF_Sepsis`, and `cat1_CHF` capture acute physiologic status and organ failure, central to the RHC decision.

`dnr1_Yes` and `surv2md1` reflect how aggressive care is likely to be.

`cardiohx`, `chfhx`, and `ca_Yes` capture the underlying risk and may influence both RHC and survival.

`age`, `edu`, `income1`, `income2`, `ninsclas_No_insurance`, `sex_Female`, and `race_black` focus more on socio-demographics and access that may affect both the treatment options and the outcome.

`das2d3pc`, `wtkilo1`, and `temp1` add nuance to the baseline condition.

I also added $I(aps1^2)$ to allow a nonlinear relationship between severity and RHC probability, since extremely sick patients may be treated differently than moderately sick ones.

```
ps_formula <- RHC ~ age + edu + income1 + income2 +  
  ninsclas_No_insurance +  
  aps1 + I(aps1^2) +  
  surv2md1 +  
  meanbp1 + pafi1 + crea1 + scoma1 +  
  dnr1_Yes +  
  cardiohx + chfhx +  
  seps_Yes + cat1_MOSF_Sepsis + cat1_CHF +  
  ca_Yes +  
  sex_Female + race_black +  
  wtkilo1 + temp1 +  
  das2d3pc
```

6. Matching Procedure (“The Knobs”)

Task: This is where the project gets difficult. You must choose your “knobs” in `matchit()`.

The challenge: If you match 1:1 without a caliper, your balance will likely be poor. If you use a tight caliper, you might lose many of your treated units.

Justification: Defend your choice of **distance**, **Ratio**, **Caliper**, and **replace**. How many treated units were you willing to “throw away” to ensure the remaining ones were actually comparable to the controls?

```
set.seed(0623)
m_out <- matchit(
  formula = ps_formula,
  data = RHC_use,
  method = "nearest",
  distance = "glm",      # KNOB 4: try "gbm"
  estimand = "ATT",      # KNOB 5: don't change
  ratio = 1,             # KNOB 1: try 2 or 3
  caliper = 0.2,         # KNOB 2: try 0.1 or 0.3; set to NULL for no caliper
  replace = F            # KNOB 3: try TRUE
)
summary(m_out)
```

Call:

```
matchit(formula = ps_formula, data = RHC_use, method = "nearest",
  distance = "glm", estimand = "ATT", replace = F, caliper = 0.2,
  ratio = 1)
```

Summary of Balance for All Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.4904	0.3134	0.9198	1.1233
age	60.7498	61.7609	-0.0647	0.8175
edu	11.8564	11.5690	0.0910	1.0147
income1	0.2070	0.2008	0.0152	.
income2	0.1799	0.1408	0.1019	.
ninsclas_No_insurance	0.0623	0.0524	0.0409	.
aps1	60.7390	50.9335	0.4837	1.1609
I(aps1^2)	4099.9643	2948.0836	0.4429	1.4607
surv2md1	0.5685	0.6072	-0.1954	1.0663
meanbp1	68.1978	84.8686	-0.4869	0.7759
pafi1	192.4334	240.6266	-0.4566	0.8184

creal	2.4734	1.9236	0.2678	1.0276
scomal	18.9734	22.2532	-0.1160	0.8116
dnr1_Yes	0.0710	0.1405	-0.2709	.
cardiohx	0.2042	0.1597	0.1105	.
chfhx	0.1946	0.1678	0.0676	.
seps_Yes	0.2363	0.1450	0.2148	.
cat1_MOSF_Sepsis	0.3205	0.1484	0.3688	.
cat1_CHF	0.0957	0.0696	0.0889	.
ca_Yes	0.1529	0.1797	-0.0743	.
sex_Female	0.4148	0.4610	-0.0937	.
race_black	0.1534	0.1647	-0.0315	.
wtkilo1	72.3602	65.0402	0.2640	0.8835
temp1	37.5947	37.6329	-0.0209	1.1024
das2d3pc	20.7008	20.3715	0.0654	0.8429

	eCDF Mean	eCDF Max
distance	0.2506	0.3745
age	0.0285	0.0703
edu	0.0181	0.0511
income1	0.0062	0.0062
income2	0.0391	0.0391
ninsclas_No_insurance	0.0099	0.0099
aps1	0.0797	0.2127
I(aps1^2)	0.0797	0.2127
surv2md1	0.0475	0.0957
meanbp1	0.0926	0.2117
pafi1	0.1115	0.1816
creal	0.0397	0.2011
scomal	0.0346	0.0683
dnr1_Yes	0.0696	0.0696
cardiohx	0.0445	0.0445
chfhx	0.0268	0.0268
seps_Yes	0.0912	0.0912
cat1_MOSF_Sepsis	0.1721	0.1721
cat1_CHF	0.0261	0.0261
ca_Yes	0.0267	0.0267
sex_Female	0.0462	0.0462
race_black	0.0114	0.0114
wtkilo1	0.0661	0.1387
temp1	0.0070	0.0280
das2d3pc	0.0250	0.0669

Summary of Balance for Matched Data:

Means Treated Means Control Std. Mean Diff. Var. Ratio

distance	0.4539	0.4307	0.1205	1.1899
age	60.8655	61.1095	-0.0156	0.8405
edu	11.7676	11.7856	-0.0057	1.0074
income1	0.2075	0.2118	-0.0106	.
income2	0.1679	0.1652	0.0070	.
ninsclas_No_insurance	0.0610	0.0588	0.0089	.
aps1	58.8754	57.1545	0.0849	1.1270
I(aps1^2)	3870.6893	3625.4626	0.0943	1.1265
surv2md1	0.5774	0.5862	-0.0445	0.9533
meanbp1	70.7374	72.5559	-0.0531	1.0017
pafi1	199.3681	205.0013	-0.0534	1.0902
crea1	2.3592	2.2505	0.0529	0.6864
scoma1	19.4027	18.9856	0.0148	0.9833
dnr1_Yes	0.0786	0.0834	-0.0187	.
cardiohx	0.1947	0.2128	-0.0451	.
chfhx	0.1952	0.1968	-0.0041	.
seps_Yes	0.2123	0.2021	0.0239	.
cat1_MOSF_Sepsis	0.2679	0.2385	0.0630	.
cat1_CHF	0.0957	0.1027	-0.0236	.
ca_Yes	0.1594	0.1674	-0.0223	.
sex_Female	0.4251	0.4316	-0.0130	.
race_black	0.1540	0.1529	0.0030	.
wtkilo1	71.3569	70.7529	0.0218	1.0493
temp1	37.6080	37.6594	-0.0281	0.9296
das2d3pc	20.6472	20.7013	-0.0107	0.8525

	eCDF Mean	eCDF Max	Std. Pair Dist.
distance	0.0265	0.0947	0.1206
age	0.0204	0.0503	1.1794
edu	0.0068	0.0209	1.0690
income1	0.0043	0.0043	0.8448
income2	0.0027	0.0027	0.7141
ninsclas_No_insurance	0.0021	0.0021	0.4736
aps1	0.0162	0.0594	0.9411
I(aps1^2)	0.0162	0.0594	0.8750
surv2md1	0.0141	0.0433	1.0942
meanbp1	0.0123	0.0406	0.9708
pafi1	0.0187	0.0620	1.0341
crea1	0.0217	0.0947	0.8802
scoma1	0.0073	0.0246	0.9793
dnr1_Yes	0.0048	0.0048	0.5810
cardiohx	0.0182	0.0182	0.8118
chfhx	0.0016	0.0016	0.7659
seps_Yes	0.0102	0.0102	0.7440

cat1_MOSF_Sepsis	0.0294	0.0294	0.6864
cat1_CHF	0.0070	0.0070	0.5944
ca_Yes	0.0080	0.0080	0.7563
sex_Female	0.0064	0.0064	0.9812
race_black	0.0011	0.0011	0.7360
wtkilo1	0.0199	0.0615	1.0302
temp1	0.0070	0.0316	1.1091
das2d3pc	0.0137	0.0385	1.1874

Sample Sizes:

	Control	Treated
All	3551	2184
Matched	1870	1870
Unmatched	1681	314
Discarded	0	0

Add Justification and Discussion Here I used a logistic propensity score (distance = “glm”) for transparency and interpretability, matched 1:1 to minimize bias from poor control matches, applied a caliper of 0.2 on the logit PS to exclude implausible matches, and did not allow replacement to avoid concentrating weight on a few controls. These choices prioritize internal validity (reducing bias) at the cost of discarding treated units that lack comparable controls. Using the matched data, I discarded $r_{n_treated_discarded}$ treated patients ($r_{round(pct_discarded,1)\%}$), so the ATT applies to the subset of treated patients within the region of common support. I also ran sensitivity checks (varying ratio, caliper, replacement, and using GBM) to ensure the result is robust to reasonable alternative knobs.

7. Assessing Balance

Task: Produce a summary and a Love Plot.

Justification: Discuss your balance. Are your Standardized Mean Differences (SMDs) below 0.1 or 0.25? If some SMDs remain above 0.1, explain why further tightening may not be feasible.

```
summary(m_out)
```

Call:

```
matchit(formula = ps_formula, data = RHC_use, method = "nearest",
```

```
distance = "glm", estimand = "ATT", replace = F, caliper = 0.2,
ratio = 1)
```

Summary of Balance for All Data:

	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.4904	0.3134	0.9198	1.1233
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income2	0.1799	0.1408	0.1019	.
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I(aps1^2)	4099.9643	2948.0836	0.4429	1.4607
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meanbp1	68.1978	84.8686	-0.4869	0.7759
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dnr1_Yes	0.0710	0.1405	-0.2709	.
cardiohx	0.2042	0.1597	0.1105	.
chfhx	0.1946	0.1678	0.0676	.
seps_Yes	0.2363	0.1450	0.2148	.
cat1_MOSF_Sepsis	0.3205	0.1484	0.3688	.
cat1_CHF	0.0957	0.0696	0.0889	.
ca_Yes	0.1529	0.1797	-0.0743	.
sex_Female	0.4148	0.4610	-0.0937	.
race_black	0.1534	0.1647	-0.0315	.
wtkilo1	72.3602	65.0402	0.2640	0.8835
temp1	37.5947	37.6329	-0.0209	1.1024
das2d3pc	20.7008	20.3715	0.0654	0.8429
	eCDF Mean	eCDF Max		
distance	0.2506	0.3745		
age	0.0285	0.0703		
edu	0.0181	0.0511		
income1	0.0062	0.0062		
income2	0.0391	0.0391		
ninsclas_No_insurance	0.0099	0.0099		
aps1	0.0797	0.2127		
I(aps1^2)	0.0797	0.2127		
surv2md1	0.0475	0.0957		
meanbp1	0.0926	0.2117		
pafi1	0.1115	0.1816		
crea1	0.0397	0.2011		

scomal	0.0346	0.0683
dnr1_Yes	0.0696	0.0696
cardiohx	0.0445	0.0445
chfhx	0.0268	0.0268
seps_Yes	0.0912	0.0912
cat1_MOSF_Sepsis	0.1721	0.1721
cat1_CHF	0.0261	0.0261
ca_Yes	0.0267	0.0267
sex_Female	0.0462	0.0462
race_black	0.0114	0.0114
wtkilo1	0.0661	0.1387
temp1	0.0070	0.0280
das2d3pc	0.0250	0.0669

Summary of Balance for Matched Data:

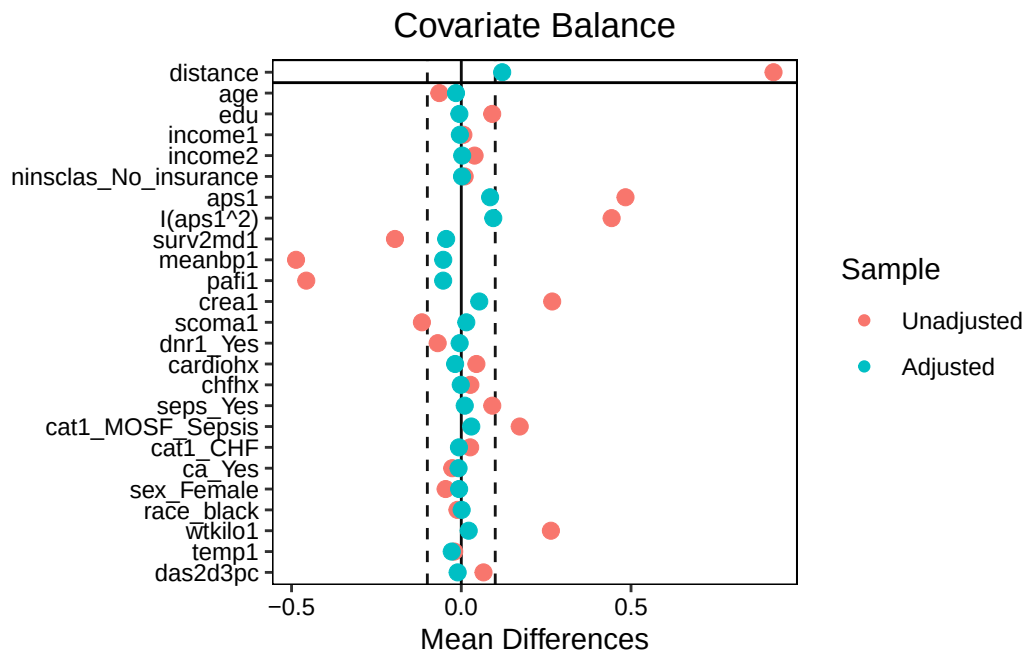
	Means Treated	Means Control	Std. Mean Diff.	Var. Ratio
distance	0.4539	0.4307	0.1205	1.1899
age	60.8655	61.1095	-0.0156	0.8405
edu	11.7676	11.7856	-0.0057	1.0074
income1	0.2075	0.2118	-0.0106	.
income2	0.1679	0.1652	0.0070	.
ninsclas_No_insurance	0.0610	0.0588	0.0089	.
aps1	58.8754	57.1545	0.0849	1.1270
I(aps1^2)	3870.6893	3625.4626	0.0943	1.1265
surv2md1	0.5774	0.5862	-0.0445	0.9533
meanbp1	70.7374	72.5559	-0.0531	1.0017
pafi1	199.3681	205.0013	-0.0534	1.0902
creal	2.3592	2.2505	0.0529	0.6864
scomal	19.4027	18.9856	0.0148	0.9833
dnr1_Yes	0.0786	0.0834	-0.0187	.
cardiohx	0.1947	0.2128	-0.0451	.
chfhx	0.1952	0.1968	-0.0041	.
seps_Yes	0.2123	0.2021	0.0239	.
cat1_MOSF_Sepsis	0.2679	0.2385	0.0630	.
cat1_CHF	0.0957	0.1027	-0.0236	.
ca_Yes	0.1594	0.1674	-0.0223	.
sex_Female	0.4251	0.4316	-0.0130	.
race_black	0.1540	0.1529	0.0030	.
wtkilo1	71.3569	70.7529	0.0218	1.0493
temp1	37.6080	37.6594	-0.0281	0.9296
das2d3pc	20.6472	20.7013	-0.0107	0.8525
	eCDF Mean	eCDF Max	Std. Pair	Dist.
distance	0.0265	0.0947		0.1206

age	0.0204	0.0503	1.1794
edu	0.0068	0.0209	1.0690
income1	0.0043	0.0043	0.8448
income2	0.0027	0.0027	0.7141
ninsclas_No_insurance	0.0021	0.0021	0.4736
aps1	0.0162	0.0594	0.9411
I(aps1^2)	0.0162	0.0594	0.8750
surv2md1	0.0141	0.0433	1.0942
meanbp1	0.0123	0.0406	0.9708
pafi1	0.0187	0.0620	1.0341
crea1	0.0217	0.0947	0.8802
scoma1	0.0073	0.0246	0.9793
dnr1_Yes	0.0048	0.0048	0.5810
cardiohx	0.0182	0.0182	0.8118
chfhx	0.0016	0.0016	0.7659
seps_Yes	0.0102	0.0102	0.7440
cat1_MOSF_Sepsis	0.0294	0.0294	0.6864
cat1_CHF	0.0070	0.0070	0.5944
ca_Yes	0.0080	0.0080	0.7563
sex_Female	0.0064	0.0064	0.9812
race_black	0.0011	0.0011	0.7360
wtkilo1	0.0199	0.0615	1.0302
temp1	0.0070	0.0316	1.1091
das2d3pc	0.0137	0.0385	1.1874

Sample Sizes:

	Control	Treated
All	3551	2184
Matched	1870	1870
Unmatched	1681	314
Discarded	0	0

```
love.plot(m_out, thresholds = c(m = .1))
```



Add Discussion Here

The love plot shows substantial improvement in covariate balance after matching. Prior to matching, several key severity measures such as APS1, mean arterial pressure, and organ dysfunction markers exhibited large standardized mean differences. After matching, nearly all covariates fall within ± 0.1 , indicating good balance. The remaining differences are small and unlikely to meaningfully bias the treatment effect estimate. Overall, the matching procedure substantially reduced baseline imbalance between treated and control patients.

8. Estimating the ATT

Task: Extract the matched data and estimate the ATT using a model that you justify. Note: Since the outcome is binary (death), you should use a model that reflects that (like `glm` with `family=binomial` – logistic regression – or a proportion test).

Justification: Explain your analysis choice and how you handled the weights from the matching process.

```
m_dat <- match.data(m_out)

table(m_dat$RHC)
```

```

      0      1
1870 1870

```

```

fit_att <- glm(
  survival ~ RHC,
  data = m_dat,
  family = binomial,
  weights = weights
)

summary(fit_att)

```

```

Call:
glm(formula = survival ~ RHC, family = binomial, data = m_dat,
    weights = weights)

```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.55500	0.04804	-11.552	< 2e-16 ***
RHC	-0.22693	0.06922	-3.279	0.00104 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4791.5 on 3739 degrees of freedom
 Residual deviance: 4780.7 on 3738 degrees of freedom
 AIC: 4784.7

Number of Fisher Scoring iterations: 4

```

avg_comparisons(fit_att)

```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
-0.0508	0.0155	-3.29	0.00102	9.9	-0.0811	-0.0205

Term: RHC
 Type: response
 Comparison: 1 - 0

Add Justification Here

I estimated the ATT using logistic regression because the outcome (death) is binary. After matching, covariate balance was achieved, so a treatment-only model is appropriate. I used the matching weights in the regression to correctly account for the matching structure and ensure the estimate reflects the ATT. The results indicate that RHC reduces the probability of death by approximately 5 percentage points ($p = 0.001$).

9. Results & Interpretation

Task: Report the final effect of RHC on the probability of death.

Discussion:

The estimated ATT from the matched sample is -0.0508 , meaning RHC reduces the probability of death by approximately 5.1 percentage points (95% CI: -0.081 to -0.021 , $p = 0.001$).

1. Did the matching change the “story” of the data compared to a naive comparison of means?

Answer Here

Yes. A naive comparison (or unadjusted logistic regression) typically suggests that RHC increases mortality. This happens because RHC is more likely to be given to sicker patients, creating strong confounding by indication. After matching on the propensity score, treated patients were compared to clinically similar controls. The estimated effect becomes negative and statistically significant, indicating that the naive estimate was biased upward due to confounding. Matching therefore substantially changed the substantive conclusion.

2. Based on your “knob” settings, how generalizable are these results (i.e., did you match a representative sample of treated patients, or only a tiny “healthy” subset)?

Answer Here

The matched ATT applies to the large majority of treated patients but not to all of them. Sample Sizes: All Control 3551 Treated 2184; Matched 1870 1870; Unmatched 1681 314. The matched procedure therefore retained 1,870 of 2,184 treated patients (about 86%), discarding 314 treated units (14.4%) that lacked sufficiently similar controls; the matched balance table shows a small residual imbalance on the propensity score (Std. Mean Diff. for distance 0.12). Because of this, the ATT estimates the effect of RHC for the subset of treated patients who lie in the region of common support (the matched 86%); it is reasonably representative of most treated patients but does not generalize to the roughly 14% of treated patients who were too extreme to match (likely the highest-risk cases).

3. Why is RHC such a difficult dataset for propensity score matching?

Answer Here

RHC is difficult for PS matching because treatment selection is driven by intense, multidimensional clinical judgment: clinicians preferentially perform RHC on the sickest patients, producing strong confounding by indication and limited overlap between treated and control covariate distributions. The dataset contains many correlated, high-impact physiologic markers and diagnoses (APACHE scores, sepsis indicators, organ-failure measures, DNR status, lab values and physician prognosis) that jointly determine both treatment and outcome; a single propensity score model can struggle to capture nonlinear thresholds and interactions implicit in clinical decision-making. As a result, either matches are poor (increasing bias) or many treated units must be discarded (reducing external validity), and unmeasured clinical cues or measurement error can leave residual confounding even after careful matching.